

Process Characterization Plan

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCP-006

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

Approvals	3
Abbreviations	3
1 Purpose and scope	5
2 Objectives	6
3 Related documents	7
4 Prior knowledge and quality risk basis	9
4.1 Unit-operation description and prior knowledge	9
4.2 Quality attributes in scope	10
4.3 Risk-based prioritization of parameters	10
5 Materials and methods	12
5.1 Scale-down model and its qualification	12
5.2 Operation	12
5.3 Analytical methods	13
5.4 Sampling plan	13
5.5 Statistical methods	14
6 Study design	15
6.1 Factors, ranges and study type	15
6.2 Screening design	16
6.3 Response-surface design	16
6.4 Univariate assessment	16
7 Acceptance and decision criteria	18
8 Proven acceptable ranges (planned analysis)	20
9 Data management and integrity	22
10 Roles and responsibilities	23
11 Deliverables and schedule	24
12 Risks and assumptions	25
13 Approvals	26
14 Appendices A-B — Planned design matrices	27
14.1 Appendix A — Planned screening design matrix	27

14.2 Appendix B — Planned response-surface design matrix	27
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References 29

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CCD, central composite

design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PCP, process characterization plan; PCR, process characterization report; PPQ, process performance qualification; RSM, response surface methodology; SD, standard deviation; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; TCID₅₀, fifty-percent tissue culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This plan defines the process characterization study for the low-pH viral inactivation step of the A-Mab drug-substance process. The step is a hold on the Protein A eluate under acidic conditions, and it is the point in the purification train at which enveloped virus is inactivated instead of being removed. The study will quantify how the hold conditions govern virus inactivation and product quality, and it will establish the operating ranges that the commercial control strategy applies at the receiving site, so that the ranges carried into commercial manufacture are the ranges this study has characterized. It is a Stage 1 activity in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), and it follows the enhanced development approach of ICH Q8(R2) (International Council for Harmonisation 2009).

The scope runs from the point at which the Protein A eluate is titrated to the inactivation pH to the point at which the held pool is neutralized for cation exchange. The step has 4 parameters in scope, of which 3 will be studied in a multivariate design and 1 in a univariate assessment, on the basis set out in §4.3. Every run of the multivariate design will be measured for 3 responses, namely the XMuLV log reduction factor, the aggregate content of the pool and the acidic charge variants of the pool.

Three things sit outside this plan. The clearance contributed by the anion exchange and small-virus retentive filtration steps is characterized in PCP-008 and PCP-009 and is not evaluated here, although the clearance credited to those steps enters this plan as a fixed input to the acceptance criterion of §7. The cumulative viral clearance claim for the drug substance is consolidated in PCMR-001, and no part of that claim is established by this step alone. Commercial-scale process performance qualification is a Stage 2 activity under PTP-001.

2 Objectives

The study has five objectives.

- (i) Quantify the effect of inactivation pH, hold time and temperature on the XMuLV log reduction factor, on pool aggregate and on pool acidic variants, over the ranges given in §6.
- (ii) Define the multivariate region in which the log reduction factor meets the step criterion of §7 while aggregate and acidic variants stay inside their drug-substance specifications.
- (iii) Establish a proven acceptable range for each parameter and each governed attribute, both at the set-points of the other parameters and with the other parameters varying inside their normal operating ranges (§8).
- (iv) Provide the evidence from which each parameter will be classified in PCR-006 under SOP-4001.
- (v) Justify the Stage 2 operating ranges for the step and its contribution to the control strategy for the drug substance.

This plan states its criteria before any data exist. No operating region, no proven acceptable range and no parameter classification is asserted here, and every outcome is reported in PCR-006.

3 Related documents

This plan sits inside the A-Mab characterization package. Its parents are the transfer plan, the pre-characterization risk assessment and the master plan, and the executed study is reported in the paired document PCR-006. The cross-references are given in Table 3.1.

Table 3.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-006	Process Characterization Report (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The study is governed by the controlled procedures and validated analytical methods in Table 3.2. SOP-1001 governs the qualification of the scale-down model, SOP-1002 the design and the analysis of the experiments, SOP-2009 the operation of the step, and SOP-4001 the classification of the parameters once the study has reported.

Table 3.2: Controlled procedures and validated analytical methods governing the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Reference	Title	Type
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Low-pH viral inactivation is the second step of the A-Mab purification train and the first step with a dedicated viral-safety function. The Protein A eluate is already acidic, so the pool is titrated to the inactivation pH, held for a defined time under temperature control, and then titrated to the load condition of the cation exchange step, which places the hold between two chromatography operations and gives it no separating function of its own. Inactivation is a chemical process and not a separation. Acid disrupts the lipid envelope and the envelope glycoproteins of susceptible viruses, so infectivity is lost while the virus particle stays in the pool. Non-enveloped viruses have no envelope to disrupt, and this step makes no clearance claim for minute virus of mice or for any other non-enveloped model virus. The claim that is made covers the enveloped model virus (xenotropic murine leukaemia virus, XMuLV) and is framed by ICH Q5A(R2) (International Council for Harmonisation 2023a).

The low-pH hold is the most established operation in the purification train. Three earlier humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) were manufactured with the same hold, and its process conditions have remained essentially unchanged for those products and through A-Mab clinical and engineering manufacture (CMC Biotech Working Group 2009). The behaviour of the step is understood mechanistically. Inactivation is faster at lower pH, at longer hold and at higher temperature, so the worst case for viral safety is the top of the pH range combined with the shortest hold at the lowest temperature, and that corner is the one the design has to resolve most precisely. Product quality moves in the opposite direction. The aggregate content of the pool rises with the length and the temperature of the hold, and acid-induced conformational change of the antibody adds a further aggregation risk at the low end of the pH scale. The two mechanisms bound the usable pH window from opposite sides.

The inactivation kinetics established on the prior products transfer directly to A-Mab, and the low-pH hold is treated here as a settled platform operation. What the study adds is the bounds of those kinetics over the ranges the commercial process will use. The commercial control strategy needs the edges of the usable window, so each parameter is expanded beyond its routine control range and the design is loaded toward those edges. The design is centred between the edges of the characterization ranges, so the study resolves the region as a whole and not only the neighbourhood of the set-point.

The material entering the step is defined by the capture step. Aggregate and charge variants in the Protein A pool are formed upstream and are not controlled here, so the

study measures both the level in the pool and the change across the hold. Aggregate is reduced downstream by cation exchange (PCP-007), and the charge-variant profile of the drug substance is set in the production bioreactor (PCP-003).

4.2 Quality attributes in scope

One drug-substance critical quality attribute is set by this step, and it is given with its acceptance criterion and its criticality in Table 4.1.

Table 4.1: Drug-substance critical quality attribute set by the low-pH hold.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20

Enveloped-virus clearance is of very high criticality, and its acceptance criterion is cumulative over the purification train instead of specific to any one step, so the criterion this study must meet is the step's own contribution to that total, which is derived in §7. Criticality is treated as a continuum under ICH Q9(R1) (International Council for Harmonisation 2023b), and the Tool #1 score in Table 4.1 combines the impact of the attribute on safety and efficacy with the uncertainty in that impact.

Two further attributes are governed by the step without being set by it (Table 4.2). Neither is cleared here. Both rise slowly during an acidic hold, so the decision rule for each is that the pool stays inside the drug-substance specification and that the change across the hold is characterized and bounded.

Table 4.2: Attributes the step must not degrade, with their drug-substance acceptance criteria.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4

4.3 Risk-based prioritization of parameters

RA-001 ranked the parameters of this step before any characterization was run, and its output is the study type in Table 6.1. The ranking combined the potential impact of a parameter on a critical quality attribute with its potential to interact with the other parameters, and a parameter with no data or rationale behind it was ranked at the highest level (CMC Biotech Working Group 2009).

Inactivation pH carries the highest rank at this step. Its consequence is two-sided. Above the top of the studied range the antibody is stable but inactivation is incomplete, and below the bottom of it the kinetics are fast but the product is at risk of aggregating. A parameter whose failure modes lie on both sides of its range has no safe direction in which to drift, which is why it is carried into the multivariate design and why the criteria of §7 are applied as a pair and not one at a time.

Hold time and temperature act on the same inactivation kinetics as pH, so their effects cannot be resolved one parameter at a time, and both are carried into the multivariate design with pH. A-Mab concentration is ranked lower. Prior knowledge indicates that protein concentration has little or no effect on inactivation kinetics or on aggregation over the range the capture step delivers (CMC Biotech Working Group 2009), and the concentration of the Protein A pool is an outcome of that step and not a set-point of this one. Its normal operating range and its characterization range therefore coincide (Table 6.1), and it will be assessed univariately as described in §6.4.

5 Materials and methods

5.1 Scale-down model and its qualification

Virus cannot be introduced into a manufacturing facility, so the clearance study will be executed on a scaled-down model of the hold with spiked virus. ICH Q5A(R2) requires that such a model be shown to represent the production step before its results are applied to that step (International Council for Harmonisation 2023a). The qualification will be executed and documented under SOP-1001, and its status will be stated in PCR-006 before any characterization result is reported.

The model matches the commercial step on the variables that govern inactivation. Control of pH, hold temperature, hold time, mixing and the surface-to-volume ratio of the vessel are matched or bracketed, and the titration profile is reproduced so that the pool reaches the target pH over a comparable time, since the titration itself is part of the exposure the virus receives. Scale-independent variables are operated at the values proposed for commercial manufacture, and scale-dependent variables are set so that the small-scale hold reproduces the performance of the full-scale hold. Qualification compares the pH trajectory after titration, the temperature profile over the hold, and the change in aggregate and charge variants across the hold, against the corresponding at-scale data from clinical and engineering batches. The comparison and its acceptance criteria are held in the qualification record under SOP-1001.

The spiking arrangement is part of the model and is qualified with it. Virus stock will be added to the load in a proportion small enough that the composition of the load is not materially altered, and the spiked load will be sampled before titration to fix the input titre of that run. Cytotoxicity and interference controls will be run on every matrix presented to the infectivity assay, and hold samples will be neutralized on withdrawal so that inactivation does not continue in the sample. A spike large enough to change the load, or a matrix that suppresses the assay, would bias the log reduction factor upward, so these controls are what bound that risk and their results will be reported with the clearance data in PCR-006.

Replicated centre-point runs are the source of the pure-error estimate used in the lack-of-fit test (§5.5). The screening design carries 3 centre points and the response-surface design carries 4.

5.2 Operation

The step will be operated at small scale in the sequence used at commercial scale, under SOP-2009. The spiked load is titrated with acid to the inactivation pH, mixed to

homogeneity, and held at the target temperature for the target time. The pool is then titrated to the load condition of the cation exchange step and filtered. Samples are drawn as described in §5.4.

Inputs controlled to platform specification are not characterized in this study. Buffer and titrant composition, vessel material, mixing rate and filter type are fixed at their platform settings, are covered by identity and quality control, and are held constant across every run of the design. Varying them would confound the parameters the study is built to resolve.

5.3 Analytical methods

The validated methods that carry the study are listed with the other controlled documents in Table 3.2. XMuLV infectivity is measured as a fifty-percent tissue culture infectious dose (TCID₅₀) under AMV-3017, aggregate by size-exclusion chromatography under AMV-3011, and acidic charge variants by imaged capillary isoelectric focusing under AMV-3013, and each method is validated for the neutralized hold pool as well as for the load presented to the step.

The log reduction factor is the difference between the log₁₀ total infectivity of the spiked load and the log₁₀ total infectivity of the held pool, with the volume of each taken into account. Method performance, including the limit of detection that bounds the largest reduction the assay can report, is held in the validation records named in Table 3.2 and is summarized in PCR-006. Where a pool sample gives no detectable virus, the reduction will be reported as at least the value calculated from the detection limit, and that run will not be treated as a point estimate of the reduction achieved.

5.4 Sampling plan

Every run gives a spiked load sample and an end-of-hold pool sample, and both are assayed for infectivity, for aggregate and for charge variants. The load sample fixes the input titre and the input quality of the run, so each run carries its own control and the reduction is calculated within the run. Load attributes (aggregate, acidic variants and protein concentration) are recorded for every run, so that a load outside its usual composition can be identified afterwards.

Time-course samples will be drawn at intermediate points during the hold on the centre-point runs. These samples confirm the shape of the inactivation curve and show whether the reduction is complete before the end of the shortest hold in the design, which is the question a hold-time range answers only indirectly. Samples are neutralized on withdrawal and handled under SOP-2009, and identity, time of withdrawal and analyst are recorded at the point of collection (§9).

5.5 Statistical methods

The analysis follows SOP-1002. Each parameter is coded so that the low and high edges of its characterization range take the coded levels -1 and +1 and the midpoint takes the coded level 0. Coding puts parameters measured in different units on a common scale, so that the estimated effects can be compared directly. The screening data will be fitted by ordinary least squares with all main effects and all two-factor interactions, and an effect will be reported as twice the corresponding coded coefficient. Significance is judged at $\alpha = 0.05$.

The response-surface data will be fitted with the full quadratic model in the multivariate parameters. Model adequacy will be judged on the coefficient of determination, on its adjusted and predicted forms, on the overall F test, and on an analysis of variance that partitions the residual variation into lack of fit and pure error from the replicated centre points. A model whose lack of fit is significant at $\alpha = 0.05$ will not be used to define the operating region, and both the deficiency and the action taken in response will be reported in PCR-006.

The two designs do different work. The screening design identifies which parameters and which interactions are active over the ranges studied. The response-surface model is the predictive model, and it is the basis of the operating region, of the proven acceptable ranges of §8 and of any prediction at a condition the design did not run. No design-space claim will be made from the screening fit.

Commercial-scale capability will be estimated by Monte-Carlo simulation of 2,000 batches, with the parameters drawn from their normal operating ranges and propagated through the fitted response-surface model. It will be reported in PCR-006 as a one-sided capability index against each criterion of Table 7.1. The capability estimate informs the control strategy, and this plan sets no numeric capability threshold as a condition of the study.

6 Study design

6.1 Factors, ranges and study type

The parameters of the step, their set-points, their characterization ranges, their normal operating ranges and their study type are given in Table 6.1.

Table 6.1: Parameters to be studied, with their characterization and normal operating ranges.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Inactivation pH	pH	3.5	3.2-4	3.4-3.6	multivariate
Hold time	min	90	60-180	60-120	multivariate
Temperature	°C	21	15-25	19-23	multivariate
A-Mab concentration	g/L	20	5-35	5-35	univariate

Every characterization range is wider than the range the process is controlled to. The ranges are expanded beyond the normal operating ranges so that the study describes the step at conditions the process will not normally see, which is what allows a future excursion to be assessed against data instead of against judgement at the time it occurs. The expansion is not symmetric, because the risk that each edge represents is not symmetric.

The pH range is the narrowest of the multivariate ranges in relative terms, and each of its edges is placed by a different mechanism. The upper edge is set where incomplete inactivation becomes the limiting concern, since the reduction achieved falls as pH rises. The lower edge is set above the pH at which acid-induced aggregation of the platform antibodies has been observed (CMC Biotech Working Group 2009), so that the study bounds the product-quality risk without running into it. Hold time is expanded upward only. A hold shorter than the routine minimum is not a condition the procedure permits, and the case that needs bounding is the hold extended by an interruption downstream. Temperature brackets its normal operating range on both sides, at the limits to which the jacket of the temperature-controlled hold vessel can be set (§5.2).

The letter codes used in the design matrices and in the analysis are given in Table 6.2.

Table 6.2: Factor legend for the coded design matrices.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

6.2 Screening design

The screening study is a full two-level factorial in the 3 multivariate parameters with 3 centre points, giving 11 runs. A full factorial estimates every main effect and every two-factor interaction without aliasing, so no effect the study is interested in is confounded with another, and the three-factor interaction, which prior knowledge gives no reason to expect, is the only term left in the residual. The centre points give a direct estimate of pure error and an indication of curvature. The run order will be randomized, and the planned coded matrix is given in Appendix A.

The screening study identifies which parameters and interactions are active over the ranges studied. It does not define an operating region, and no range will be justified from it alone.

6.3 Response-surface design

The response-surface study is a face-centred central composite design in the same 3 parameters, with 8 factorial runs, 6 axial runs and 4 centre points, giving 18 runs. The axial runs sit on the faces of the design cube, so the design adds the curvature terms without going outside the characterization ranges of Table 6.1. Each parameter is observed at its low edge, its midpoint and its high edge, which is what the estimate of a quadratic term requires.

Every screening parameter will be carried into the response-surface design whatever the screening outcome. The design is small, a parameter dropped after screening cannot contribute a curvature term or an interaction to the predictive model, and the cost of carrying a parameter that turns out to be inactive is a few runs against the cost of a second execution. The planned coded matrix is given in Appendix B.

6.4 Univariate assessment

A-Mab concentration will be assessed one parameter at a time, at the low, middle and high points of its characterization range, with the other parameters held at their set-points. These runs are spiked, sampled and assayed exactly as the multivariate runs are. The assessment answers a single question, which is whether the log reduction

factor and the change in aggregate across the hold are constant over the concentration range the capture step delivers.

A null result is an outcome and not a gap. If concentration has no measurable effect on either response, the parameter will be classified on that evidence in PCR-006, its full characterization range will be carried forward as an acceptable range, and the runs will remain in the knowledge space as evidence that the step is insensitive to it.

7 Acceptance and decision criteria

The study passes when the criteria of Table 7.1 are met simultaneously over a contiguous region of the multivariate parameters. Meeting them individually at different conditions is not a pass, because the process must satisfy all of them at whatever condition it is run.

Table 7.1: Acceptance criteria for the measured responses, with the basis of each.

Measured response	Basis of the criterion	Criterion applied to the pool
Aggregates (HMW, %)	Drug-substance specification	$\leq 5\%$ HMW (SEC)
XMULV LRF ($\log_{10}$)	Cumulative requirement less the clearance credited to Steps 8 and 9	$\geq 4.93 \log_{10}$ across the step
Acidic variants	Drug-substance specification	$\leq 40\%$ (CEX/icIEF)

The clearance criterion is a floor, and it is derived rather than specified. The drug-substance requirement for enveloped virus is cumulative over the purification train and is given in Table 4.1 as at least $16.7 \log_{10}$. The anion exchange and small-virus retentive filtration steps (Steps 8 and 9) are credited with defined contributions to that total in the modular claim, and subtracting those contributions from the cumulative requirement leaves what this step must deliver on its own. The resulting floor is $4.93 \log_{10}$, and every condition inside the operating region must meet it.

The clearance criterion is applied to the mean response predicted by the model. A response surface predicts mean levels, so a region drawn on the mean alone offers only even odds of meeting the criterion at its own edge. The region will therefore be reported twice, once on the mean prediction and once with a prediction interval applied, and the second of these is the region that supports the control strategy, because the control strategy has to hold for a batch and not for the average of many batches. The interval accounts for process variation, for assay variation and for the uncertainty of the fitted model itself.

The remaining criteria are ceilings taken directly from the drug-substance specifications. The step is credited with no reduction of either attribute, so the decision rule is that the pool stays inside its specification everywhere in the region and that the change across the hold is quantified against the load. An increase that remains inside the specification is an acceptable outcome, and it will be reported with its magnitude and not only with its disposition.

Model adequacy is a precondition of the operating region and not a criterion applied to the process. The region will be declared from the response-surface model only where the lack-of-fit test is not significant at $\alpha = 0.05$ and the predicted coefficient of determination supports prediction and not merely description. Where a response fails that test, the range for the parameters governing it will be set from the observed data at the design points, and the limitation will be stated in PCR-006.

Two outcomes end the study without an operating region. If no contiguous set of conditions satisfies every criterion at once, the ranges will be re-cut and the study repeated under a revised plan. If the scale-down model fails its qualification, no result from the study will be used, and the failure will be reported under SOP-1001.

Meeting the step criterion does not establish the viral safety of the drug substance. The cumulative claim is assembled in PCMR-001 from this step, from PCR-008 and from PCR-009, and the argument that the contributing mechanisms are orthogonal is made there.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range will be reported for each governed attribute and each of the multivariate parameters. The acceptance basis is the one used in §7. For aggregate and acidic variants it is the drug-substance specification, and for enveloped-virus clearance it is the back-calculated step floor of Table 7.1.

Two analyses will be run for every pair of parameter and attribute. The first holds the other parameters at their set-points and scans the parameter of interest across its characterization range. The second propagates the normal operating ranges of the other parameters, drawing each of them at random inside its own range and evaluating the fitted response-surface model at every point of the scan. The scan uses 81 grid points across the characterization range and 2,000 draws at each grid point, under a fixed random seed so that the analysis reproduces exactly.

The second analysis is the reproducible default for the control strategy. It answers the question a manufacturing deviation actually raises, which is whether one parameter is acceptable while everything else sits somewhere inside its own range instead of at a nominal value. The first analysis is retained because it isolates the parameter, and the difference between the two is a direct measure of how much the other parameters matter.

The decision rule is the same in both analyses. The proven acceptable range of a parameter is the contiguous interval around the set-point over which the predicted response stays inside its acceptance criterion, and it is reported in the units of the parameter. Where both analyses give the same interval and that interval covers the whole characterization range, the parameter is acceptable everywhere it was studied and is also insensitive to co-variation of the others inside their normal operating ranges, as far as the fitted model can show. Where the propagated interval is narrower, the attribute that binds it will be named and the residual margin to the criterion will be given.

Each governed attribute will be reported with a figure that plots the response against its governing parameter, with the acceptance limit drawn, the acceptable region shaded, and the set-point and the normal operating range marked. The two analyses are the two panels of that figure.

A proven acceptable range and an operating region answer different questions. A proven acceptable range is a statement about one parameter under stated conditions for the others. The region of §7 is multivariate. Where every proven acceptable range covers the whole characterization range, the constraint that binds routine operation

is a corner of that region and not any single parameter, and PCR-006 will say which corner it is.

9 Data management and integrity

Study data will be recorded, reviewed and retained under the site data-integrity procedures, and every record will be attributable, legible, contemporaneous, original and accurate, as well as complete, consistent, enduring and available. Raw analytical data remain in the validated systems of the analytical laboratory. The design, the run order, the fitted models and the derived ranges are held in the statistical environment identified in SOP-1002, together with the seed and the script that reproduce them.

Each dataset is reviewed by a second qualified person before it is analysed, and the analysis is checked against the raw records before PCR-006 is issued. Any correction to a recorded value carries its reason, its author and its date, and the original value stays legible.

10 Roles and responsibilities

The study is executed by Process Development at the sending site, with the analytical and virological work performed by the functions named in Table 10.1. Each function approves the part of the plan it will execute.

Table 10.1: Functions and their responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owens this plan, executes the scale-down runs, analyses the data and authors PCR-006.
Analytical Development	Performs the size-variant and charge-variant assays and maintains their validated status.
Virology laboratory	Prepares and titrates the virus stock, performs the infectivity assay and runs the cytotoxicity and interference controls.
Manufacturing Science	Confirms that the scale-down model represents the commercial step and reviews the proposed operating ranges.
Statistics	Approves the designs, the analysis and the model-adequacy assessment.
Quality Assurance	Approves this plan and PCR-006 and disposes any deviation from the plan.

A departure from this plan is raised, investigated and dispositioned under the site deviation procedure. Every deviation is reported in PCR-006 with its investigation, its root cause where one is established, and its assessed impact on the conclusions of the study.

11 Deliverables and schedule

The deliverable of this plan is PCR-006. That report gives the qualification status of the scale-down model, the screening and response-surface results, the operating region, the proven acceptable ranges, the parameter classifications and the step's contribution to the control strategy. It rolls up into PCMR-001.

The study runs in four phases. The scale-down model is qualified first, and no characterization run starts before its qualification is accepted. The screening study runs next. The response-surface study and the univariate assessment follow, and they may run in parallel where virus stock and assay capacity allow. Analysis and reporting follow the last analytical result. No calendar dates are fixed in this plan, and the campaign schedule is held under PCMP-001.

12 Risks and assumptions

The study rests on four assumptions, and each one carries a risk to the conclusions drawn from it.

- The scale-down model represents the commercial hold. This is the largest assumption in the study, because every range it justifies is applied at commercial scale, and it is managed by the qualification described in §5.1 and by the comparison against the clinical and engineering data at scale that PCR-006 will present.
- The spiked load behaves like the unspiked pool. A virus spike adds material the commercial process does not contain, and the spike proportion is held low and controlled for that reason. A residual assumption remains, and it is stated as a limitation of the clearance claim in PCR-006.
- The load entering the step is representative of routine capture material. Aggregate and charge variants are formed upstream, so a load outside its usual composition would move the pool results without any parameter of this step changing. Load attributes are recorded on every run so that the assumption can be tested after the fact.
- The fitted model holds only inside the characterization ranges. No prediction will be made outside them, and operation outside them is a change managed under the quality system and not a movement supported by this study.

Two further risks are practical. Virus stock and infectivity-assay capacity (shared with the studies of PCP-008 and PCP-009) set the rate at which runs can be executed, and a failed assay costs a repeat of the analysis and, in the worst case, of the run. Availability of representative Protein A pool material is the second constraint, and the study will use material pooled from more than one upstream batch where a single batch is insufficient. Both risks affect the schedule of the study and neither affects its conclusions.

13 Approvals

This plan takes effect when the approvers named in Table 1 have signed it. No characterization run under it may start before that date. Any change to the design, to the ranges or to the acceptance criteria requires the same approvals as the original plan, and the change is recorded before the affected runs are executed.

14 Appendices A-B — Planned design matrices

The planned design matrices are given below in coded form, with the letter codes of Table 6.2. Coded -1 and +1 are the low and high edges of each parameter's characterization range and coded 0 is its midpoint. No response column appears, because no run has yet been executed.

14.1 Appendix A — Planned screening design matrix

Table 14.1: Planned coded screening design matrix (run order to be randomized at execution).

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	center	0	0	0
10	center	0	0	0
11	center	0	0	0

14.2 Appendix B — Planned response-surface design matrix

Table 14.2: Planned coded response-surface design matrix (run order to be randomized at execution).

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1

Run	Type	A	B	C
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	axial	-1	0	0
10	axial	1	0	0
11	axial	0	-1	0
12	axial	0	1	0
13	axial	0	0	-1
14	axial	0	0	1
15	center	0	0	0
16	center	0	0	0
17	center	0	0	0
18	center	0	0	0

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